

— IQOO HACKATHON 2026 · PUNE CITY BATTLE · WORKING PROFESSIONALS

CLASHFIT

Your body is the controller.
Your camera is the referee.

Camera-verified effort becomes damage against a boss. Pose estimation, form scoring, a fatigue model and a language model all run on the phone — and nothing ever leaves it.

TEAM DA GOATS

OMKAR KADAM

UJJWAL PARDESHI

WORKING PROTOTYPE RUNNING

CLASH-FIT.VERCEL.APP

— THE PROBLEM

NOTHING ON YOUR PHONE CAN SEE THE REP.

Every fitness app counts what you **tell** it. Tap ten, it logs ten. There is no sensor in that loop — only a text field and your honesty at the exact moment you are least able to judge yourself.

WHAT THE APP KNOWS

"10 squats"

A number you typed. No depth, no range, no tempo, no alignment, no idea whether rep 9 looked anything like rep 1.

WHAT ACTUALLY HAPPENED

10 reps. 4 of them shallow.

Depth fell 4 cm across the set. Descent got faster as control went. The knee drifted inside the toe on the last three.

The gap between those two boxes is the entire product.

— HOW IT WORKS

REPS IN. DAMAGE OUT.

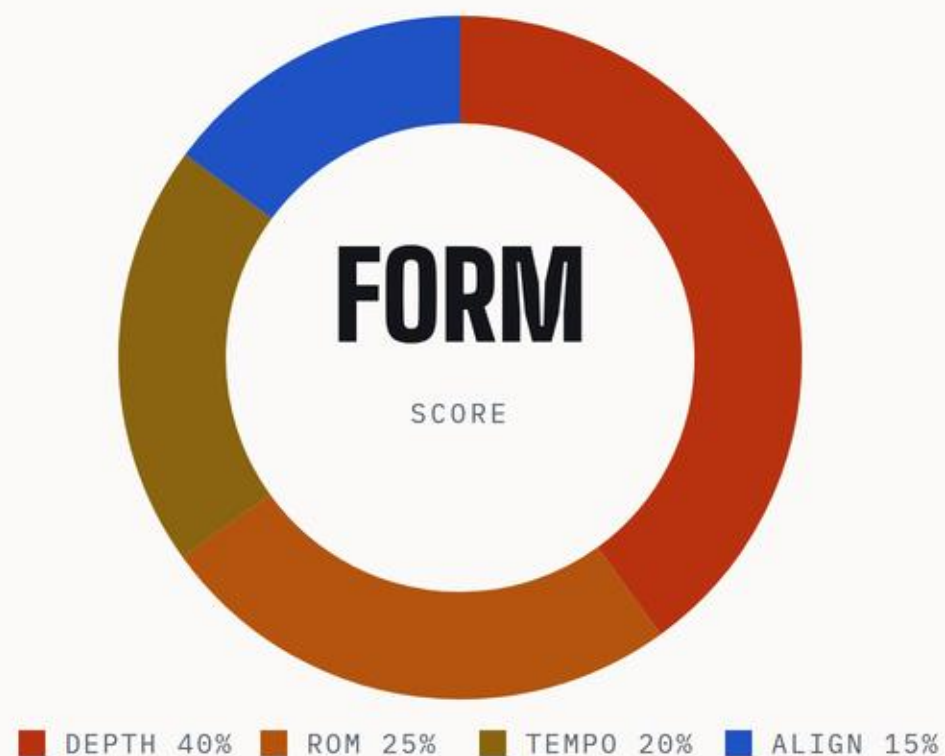
EVERY BOX BELOW RUNS ON THE PHONE. THERE IS NO SERVER IN THIS DIAGRAM.



A half-rep buys nothing. That is the whole design: the only way to do more damage is to move better, and the phone is the thing deciding whether you did.

— WHAT “GRADED” MEANS

ONE REP, FOUR MEASUREMENTS. A BAD REP VISIBLY DOES LESS.



A rep is not a count. It is four quantities the camera can measure, weighted per exercise — depth matters more in a squat than tempo does.

0.40 **Depth** in real centimetres, scored superlinearly

0.25 **Range** against your own first three reps

0.20 **Tempo** — the eccentric, where reps get cheap

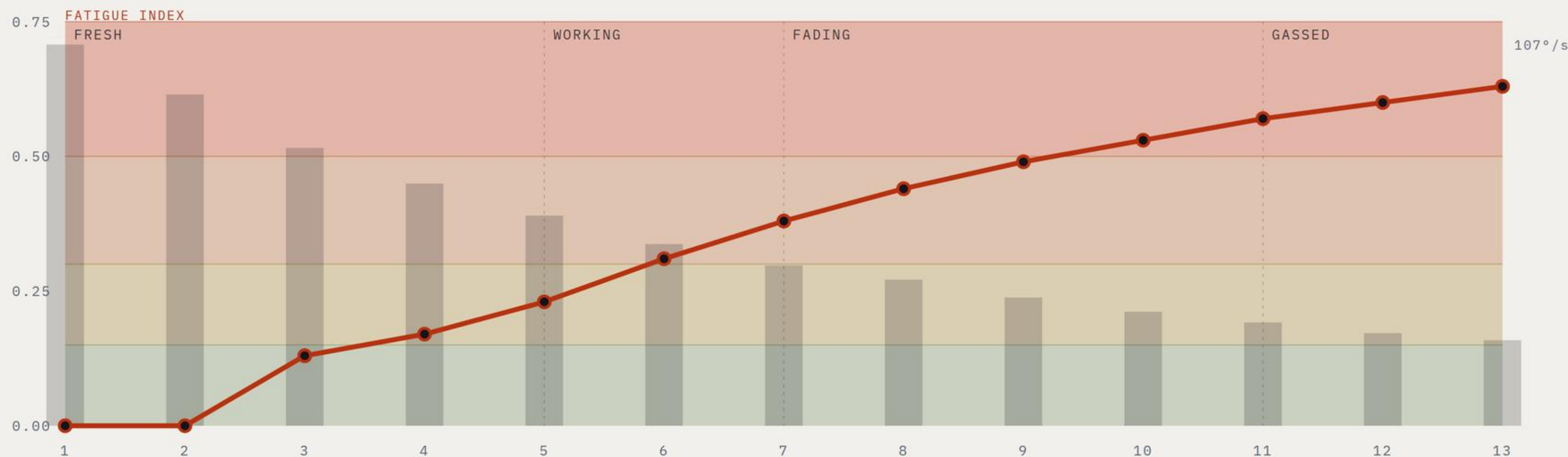
0.15 **Alignment** — knee over toe, hip above shoulder

That score becomes damage on a curve. A clean rep takes three hundred off the boss; a shallow one takes almost nothing. The only way to win is to move better.

WEIGHTS FROM CONFIG/EXERCISES/SQUAT.JSON · CURVE FROM CONFIG/COMBAT.JSON

IT KNOWS WHEN YOU'RE GASSED.

Velocity, range and the pauses between reps, measured every rep against **your own first three**. Not a difficulty slider. Not a self-reported effort rating. The point at which the movement actually changed.



A PLANK HAS NO VELOCITY. SO WE MEASURE THE SHAKE.

Fatigue is not one signal. Each family of movement degrades in its own way, so each has its own weighted signals feeding the same estimator and the same four bands. That is what lets one engine cover yoga, cardio, holds, jumps and reps.

	FAMILY	EXERCISES	GAME	HOW FATIGUE SHOWS UP
	REP_CYCLE	10	Boss Fight · strength reps	Velocity decay 0.45 · range decay 0.35 · growing pauses 0.20
	ISOMETRIC_HOLD	9	Siege · your plank is the shield	Tremor growth 0.60 · quality decay 0.40
	POSE_MATCH	14	Sigil · each asana a constellation	Accuracy drift 1.00
	CADENCE	11	Pursuit · distance is cadence	Cadence decay 0.55 · amplitude decay 0.45
	BALLISTIC	7	Breaker · jump height in centimetres	Height decay 0.60 · landing softness 0.40

51

EXERCISES

5

DETECTOR FAMILIES

1

FATIGUE ESTIMATOR

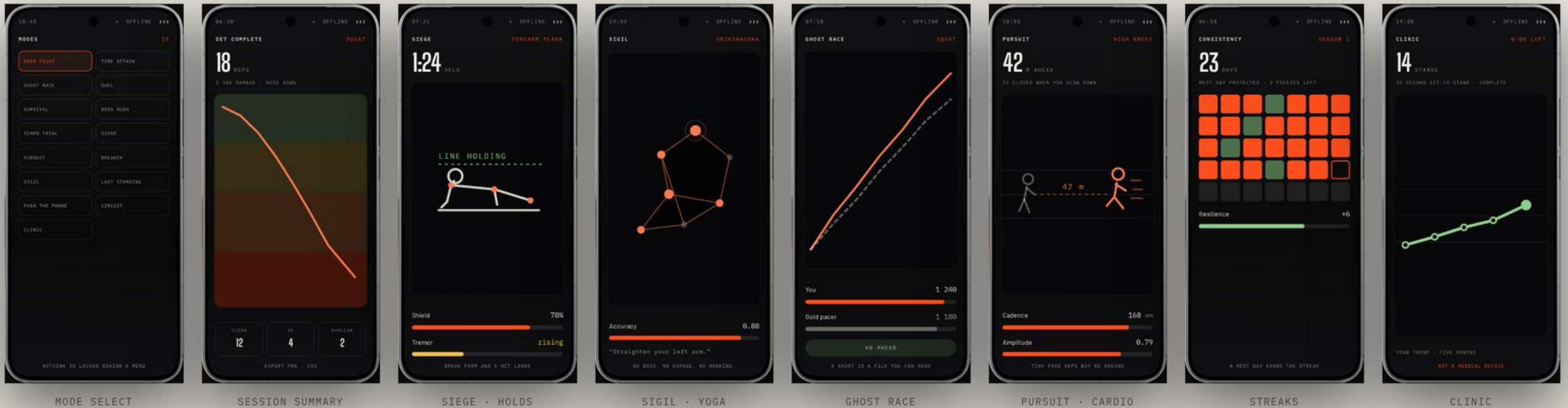
4

BANDS · 0.15 / 0.30 / 0.50

— EVERY SCREEN

ONE DEVICE. EVERY MODE.

Boss fights, holds, yoga, cardio, a race against a file, a streak and a clinical test — the same engine, the same phone, eight of the sixteen shown here.



TWO PHONES, ONE BOSS, NO SERVER.

Phones exchange **events**, not state. Boss HP is a pure function of the deduplicated union of every damage event either phone has seen, so the two screens converge without any authority deciding who is right.

$$\text{bossHp} = \text{maxHp} - \sum \text{damage over dedupe}(\text{union})$$

Every message carries a self-healing tail of the last eight events, so a dropped packet repairs itself on the next one. No acknowledgements, no retransmit logic, no clock sync — and it works in airplane mode.

DUEL · MOST DAMAGE WINS

RAID · 3-8 CO-OP

LAST STANDING · FATIGUE KO

PASS THE PHONE · ONE DEVICE

EVENTS, NOT STATE



Same boss HP on both screens

NO SERVER · NO ACKS · AIRPLANE MODE

THE COACH AND THE VILLAIN ARE THE SAME MODEL.

Between sets — never during one — a twenty-three-field summary of what your body just did goes to **Gemma 3n**, running on the phone. It writes the coaching and it writes the taunt. It never sees a frame of video, because we never send it one.

GOES TO THE MODEL

Rep count, form means, first-three vs last-three

Depth in centimetres, and how much of it you lost

Velocity loss, range loss, fatigue band

Best and worst rep, combo, boss HP, rest seconds

Bilateral deficit, and which side is weaker

NEVER GOES

Video

Images

Landmarks

Audio

Identity

EVERY FIELD ENUMERATED IN SRC/COACH.JS • SUMMARISE()

“Your last three reps lost four centimetres of depth. Take forty seconds, then finish it.”

COACH

“Your knees are negotiating. I don't negotiate.”

THE PACEMAKER

And when the model is wrong, it does not speak. A validator checks every sentence against the telemetry it was given. A line that cites a number we did not measure is dropped, and a template bank keyed on the same numbers speaks instead. The athlete never hears a hallucination.

YOUR BODY NEVER LEAVES THE PHONE.

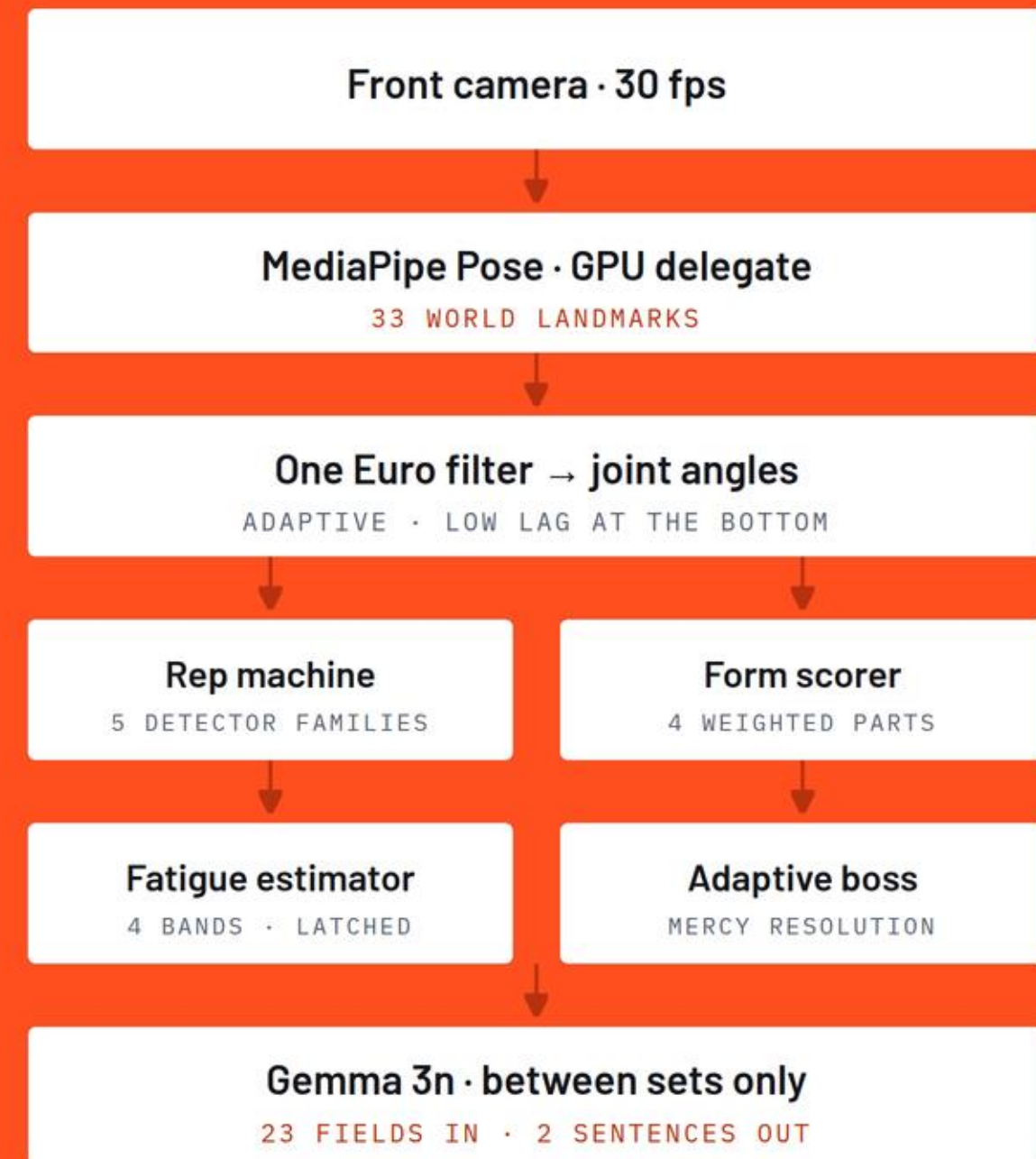
Pose, form scoring, fatigue and the language model all run on the phone. No frame of video, no landmark and no form score has ever left a device, and none can — in any mode. **Outbreak** is the single mode that uses the **network**, because a map is tiles from a server. It asks for location when you start it, never at install, and it never opens the camera.

0
FRAMES UPLOADED

0
ACCOUNTS

0
LANDMARKS SENT

0
RUNTIME DEPENDENCIES



— EVERYTHING IN THE APP

SIXTEEN WAYS TO PLAY, AND A LOT THAT IS NOT A GAME.

SIXTEEN MODES

Boss Fight	Strength reps
Time Attack	Sixty seconds
Survival	Waves
Boss Rush	Three in a row
Tempo Trial	Eyes-free
Ghost Race	Your own best
Siege	Holds
Sigil	Yoga, 14 asanas
Pursuit	Cardio
Breaker	Jumps
Circuit	All five families

TOGETHER, IN REAL TIME

Duel	Two phones
Raid	Three to eight
Team vs Team	Two bosses
Last Standing	Fatigue knockout
Pass the Phone	One device
Outbreak	Outdoors, on a map
A live leaderboard on every screen in the room, ranks and damage moving as people move. No server, no internet needed.	
CLINIC	
Sit-to-stand	Thirty seconds
Balance and reach	Monthly
Left against right	Every set

NOT A GAME AT ALL

Wake-up alarm
Stops only when the reps are done
App budget
Paid off in movement
Desk timer
Sixty seconds every fifty minutes
Posture score
Frames read, then discarded
Guided breathing
Verified, not assumed
Run tracker
Route never leaves the device
KEPT FOR YOU
Levels · XP earned from rep quality
Leaderboards global and friends, real-time
Weekly challenge same for everyone, Monday
Badges bronze, silver, gold
Session summary, exportable
Streaks that count rest as training
Fatigue history across months

— OUTDOORS • THE ONE MODE THAT GOES ONLINE

SIXTY SECONDS. THEN THEY START MOVING.

Outbreak puts the chase on a real map. A sixty-second head start, then pursuers spawn around you and begin closing. Ammo caches sit at real coordinates. Get within twelve metres of a pursuer and it has you.

60s

HEAD START

400m

SPAWN RADIUS

12m

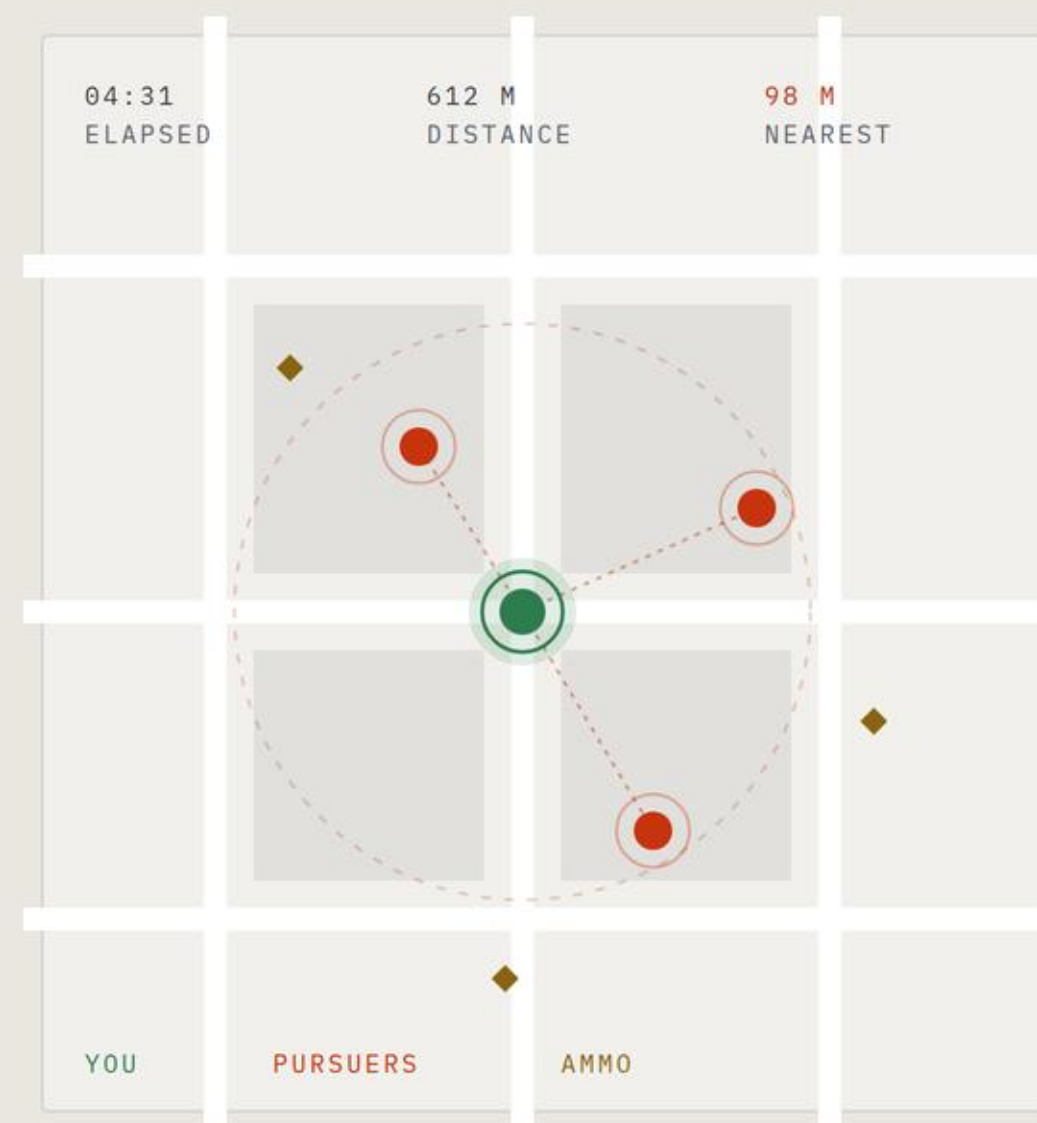
CAPTURE RADIUS

×6

AMMO CACHES

A pursuer closes when your cadence drops, not only when your coordinate stops. The same detector that runs Pursuit indoors decides whether you are still running — which is why walking it does not work.

And it is the one mode that asks for the network. A map is tiles from a server. Outbreak requests location when you start it, never at install; declining costs you exactly this mode. The camera stays shut throughout, so no frame leaves the device here either, and the route is stored on the phone.



— WHY THIS PHONE

TWO HARD JOBS, ONE BATTERY.

90°

The spec that actually decides it

The 32 MP front camera's ninety-degree field of view holds a whole standing body from about two metres — the distance a bedroom gives you. A narrower selfie lens crops your feet, and a rep the camera cannot see is never counted.

3 nm

Snapdragon 8 Elite Gen 5

Pose on the GPU delegate leaves the CPU free for the rep machine, the fatigue model and the fight inside one frame's budget — then the same silicon runs a language model between sets with the radio off.

7000

A set is not a benchmark run

Twelve minutes of continuous camera, GPU and screen is a thermal problem, not a peak one. A 7000 mAh cell and iQOO's 8K vapour-chamber cooling are what keep the frame rate from sagging halfway through and taking the fatigue baseline with it.

×2

Two of them make a duel

The multiplayer story needs a second phone and nothing else. No server, no router, offline-first.



SPECIFICATIONS CROSS-VERIFIED AGAINST IQOO'S OWN PRODUCT PAGES, GSMARENA AND THE CHIP VENDOR. NOTHING ON THIS SLIDE IS QUOTED FROM A SPEC WE COULD NOT SOURCE TWICE.

THE SAME CAMERA, DOING SERIOUS WORK.

BILATERAL ASYMMETRY

Both sides were always there

Pose gives a left knee and a right knee. Every app averages them and throws the difference away. The Limb Symmetry Index is a measure clinicians use, and no consumer fitness app reports it — none of them can see both sides.

$$\text{LSI} = \text{weaker} / \text{stronger} \times 100$$

Graded for confidence, because a side-on camera foreshortens the far limb and will invent an asymmetry that is not there.

CLINIC MODE

The tests a physio uses

The thirty-second sit-to-stand is a published, validated test of lower-limb strength — counted by a human with a stopwatch, once or twice a year. The camera counts it correctly at home, as often as you like, and plots the trend the annual version can never show.

14

STANDS IN 30 SECONDS

A measurement, not a diagnosis — and the language validator enforces that.

FIVE HEALTH TOOLS

Four never involve a boss

Alarm you cannot dismiss from bed

App budget paid off in reps

Desk timer — 60s every 50 min

Posture score — frames read, then discarded

Breathing, verified from chest motion

Nothing here uploads. Nothing here is a diagnosis.

— IT ALREADY RUNS

NOT A MOCKUP. AND NOT A SOLO PROJECT.

160

TESTS PASSING

31

MODULES

0

DEPENDENCIES

51

EXERCISES

16

MODES WIRED

5 032

LINES OF ENGINE

```
$ node test/replay.js traces/...
```

```
13 reps · fatigue 0.63 · GASSED · MERCY FIRED
```

The fatigue curve three slides back is that command's output. Tests cover the rep machine, the form scorer, the fatigue estimator, the duel sync and the page's own claims — the build fails if the site prints a number the config does not hold.

TEAM DA GOATS

Omkar Kadam

ANDROID & PERCEPTION

The camera path, the pose pipeline and the detector families — everything between a frame arriving and a rep being counted.

Ujjwal Pardeshi

GAME SYSTEMS & INFERENCE

The combat model, the multiplayer sync and the on-device language model — everything between a rep being counted and it meaning something.

Prop your phone.
Fight it out.

clash-fit.vercel.app

WORKING PROFESSIONALS

TEAM OF TWO

PUNE CITY BATTLE